

Ep. 366: Mikihiro Matsuoka on Japan's Inflation, BOJ Policy Shifts, and the Weak Yen Challenge

Podcast: Macro hive

AI Executive Summary

Core Thesis The episode argues that Japan's recent inflationary trends are structurally fragile, driven primarily by supply-side shocks rather than sustainable domestic demand. Chief Economist Matsuoka-san contends that because Japan's potential growth is exceptionally low (approx. 0.5%), traditional inflation targeting is an unsuitable and potentially harmful mandate for the Bank of Japan (BOJ). Instead, he advocates for a shift toward nominal GDP growth targeting to insulate monetary policy from supply-side volatility and suggests that structural reforms—such as promoting work-from-home (WFH) policies to boost birth rates and addressing the labor input decline—are more vital than fiscal interventions like consumption tax cuts.

Key Arguments & Data

- * **Inflation Decomposition:** While headline CPI has remained above 2%, recent increases are characterized by a lack of demand-side contribution; current inflation is driven by negative supply shocks, which is detrimental to long-term economic activity.
- * **The Neutral Rate Discrepancy:** While the BOJ has signaled a neutral rate in the 1% to 2% range, Matsuoka estimates the true nominal equilibrium rate for Japan should be much lower (potentially around 0.75%) due to the direct correlation between low potential growth and steady-state inflation.
- * **Monetary Policy Mandate:** Matsuoka proposes "Nominal GDP Growth Targeting" as an alternative to "Inflation Targeting." This would prevent the BOJ from being forced to hike rates in response to supply-side shocks (like oil price spikes) that do not reflect underlying economic strength.
- * **Wage and Labor Divergence:**
 - * Monthly wage growth is at a 30-year high of approximately 3%, providing much-needed support for household income.
 - * However, a significant gap exists between large corporations (which can absorb labor costs) and small/mid-sized companies (where personnel costs remain too high relative to sales to allow for significant raises).
 - * "Work-style reforms" and social security tax thresholds are inadvertently causing a decline in total hours worked per person, acting as a negative supply shock.
- * **Fiscal Vulnerability:** Despite concerns over Japan's massive debt, Matsuoka's proprietary "fiscal vulnerability index" ranks Japan 16th among 24 OECD countries—notably more stable than the United States. However,

he warns of "fiscal dominance," where government spending patterns may begin to dictate or override monetary policy. * **Currency Anomalies:** The Japanese Yen is currently trading at approximately four standard deviations weaker than models would predict, a movement that cannot be easily explained by interest rate differentials or balance of payments data. *

Demographic Drivers: Analysis suggests that government spending on childcare has no significant correlation with birth rates; conversely, an increase in "Work From Home" (WFH) flexibility correlates with a 0.3 to 0.5 increase in the expected number of children per lifetime. * **US**

Economic Trends: The "US premium" in equity valuations is narrowing, and the massive vertical jump in AI-related CAPEX (contributing roughly 0.7% to GDP growth recently) may be unsustainable in the long term.

Actionable Takeaways * **For Investors in Japanese Equities:** Prioritize large-cap companies over small/mid-cap firms. Large-caps have demonstrated the margin resilience necessary to sustain current wage growth, whereas smaller firms face significant margin compression from rising personnel costs. * **For Macro Strategists:** Shift focus from headline CPI to Nominal GDP trends. As the debate shifts toward nominal targeting, the sustainability of Japanese economic health will be better measured by the expansion of total nominal output rather than just price levels. * **For FX Traders:** Avoid relying solely on interest rate differentials to predict Yen movements. Given that the Yen is trading at extreme deviations from model expectations, monitor Bank of Japan intervention patterns, which are likely aimed at curbing volatility/depreciation rather than forcing appreciation. * **For Policy Analysts:** Recognize that traditional fiscal "stimulus" (like consumption tax cuts) may cause more harm than good via demand dislocation and regressive wealth distribution; focus instead on structural labor policies like WFH to address long-term demographic decline.

Full Transcript

Welcome to the MacroHive Conversations podcast. I'm Bilal Hafiz, Head of Strategy at MacroHive. Before we dive in, a quick update. Following MacroHive's acquisition by the BGC Group, BGC has launched a new regulated business called MacroHive Markets. For the first time, onboarded clients can also access execution services via MacroHive Markets. This new workflow will allow Macro institutional investors to benefit from strategy and quant insights, alongside access to global liquidity. Please contact Andrew Simon or Gail Mojibriyan

Bloomberg or click the link in the description below for more details on McRide markets and its service offering. Now, on to this episode's guest, Microhero Matsuoka. Matsuoka-san is the Chief Economist of SBI Securities in Japan. Before that, he was the Chief Economist for Japan at Deutsche Bank. And overall, he has been involved in macroeconomic analysis at research institutions and financial institutions for the past 30 years or so. He is known to be one of the leading Japan economists with unique insights on structural issues. And over the years, he has been highly ranked in numerous surveys, including the Institutional Investor Survey. Now on to our conversation.

Greetings and welcome Matsuoka-san. It's fantastic to have you back on the podcast show. It's been a few years now. Hi. Thank you for calling me back. I'm very grateful. Great. Japan has so many interesting macro topics at the moment. One of the bigger ones, I suppose, is inflation. It does appear that Japanese inflation has finally broken its deflation trap that it had. And inflation is... seems to be moving above 2% in a stable fashion, but there are lots of one-off factors here and there. So in your view, how would you characterize inflation in Japan? What are the demand factors affecting it? What are the supply factors affecting it? The headline CPI inflation is slowly above 2% for several years. Then we look at the decomposition.

of inflation to demand side factors and supply side factors where the demand side factors are the items where the price and quantity is moving in the same direction and supply side factors are the items where the quantities

and prices are moving in opposite direction and as such we look at the composition we found that the last part of the increase in the first round of the inflation in every part of this decade, mainly came from supply side factors, i.e. negative supply shock of several rounds. And since 2004, we have started to see an increase in the contribution from the demand side factors, i.e. the purchase part of the inflation, so that prices and quantities are moving upward simultaneously.

However, in the last 12 months, we've seen a declining contribution from the demand-side practicals and a reversal of supply-side contribution, i.e., currently, we are in a very inconvenient mix of composition. Very much lack of demand-side contribution and just the supply-side contribution, which is actually not good for economic activity. situation. So it does sound like you're a bit skeptical that Japan has fully escaped the deflation trap. Is that a fair characterization? I wouldn't say Japan hasn't, but I mean, it probably has an effect. However, our goal of reaching 2% is still very difficult to be maintained. Once you strip out supply-side contributions. And one other thing we'd like to focus is that there is a correlation between economies' long-run potential growth and steady state inflation, so that the higher the potential, the higher the steady state inflation. In case of Japan, our potential growth rate can be as low as a half percentage point.

compared with, say, 2% for the US as such. I think that steady-state inflation in Japan cannot be as high as 2%. That's my impression. And in terms of the short-term equilibrium interest rate or neutral interest rates, the Bank of Japan recently in their minutes or summary of opinions, mentioned 2% as a neutral rate. I mean, I think if my memory serves, that's the first time they've mentioned an explicit number in many, many years for the neutral rate. Before then, they would just talk about neutral rate is here or there, like without specifying a number. What's your sense? Is 2% the neutral rate? According to your estimates, what do you think the neutral rate should be?

Rain range, I think so previously they are talking about say middle of 1% Maybe up to 2% return half so the 2% is probably in the mid range for the estimate and It may be the case that some people would not every totally with the 2% So there is a some range of estimates, I think but generally

speaking that range is moving higher and higher every quarter or so. And I am probably in a quite minority camp in the sense, given the correlation between potential growth and steady-state inflation, i.e., the higher the economic growth, the higher long-run inflation. That's why the EM country inflation tends to be higher than the international developed countries. So I would imagine that the nominal critical interest rate in Japan may be around 1% or possibly below 1%. The way I do this calculation is very simple. Look at the US and US potential growth rate is probably about 2%. Their target of inflation is 2%.

trade funds like equilibrium is around 3% at the moment. So they are talking about about 1% real equilibrium interest rates for the economy with 2% potential growth. So there is a 1% point gap in favor of the economy growth. If you apply that to Japan with the potential growth rate of 0.5%, our real equilibrium interest rate can be as low as minus 0.5. Then on top of that, we had to add steady state inflation, which is a function of potential gross rate. We have potential gross rate in Japan is lower than the US by about 150 basis point. So I would assign coefficient of 0.5 to the potential gross rate to explain inflation. So I think On top of negative 0.5% real interest rate, we add only 1.25% point as the steady state inflation, not entirely 2%, which end up with very low 75% points. It's a bit complicated, but basically the message is short-term equilibrium interest rate.

is influenced by both the potential growth and the steady state inflation. And steady state inflation is a function of an equitable interest, sorry, the function of potential growth. So essentially, I mean, your point then is that you think the neutral rate should be lower than what the BOJ has for its mid-rate at the moment. I mean, there is a larger question around what should be the right monetary policy objective for Japan. You've had some writings around this all. The convention is that inflation should be the target for central banks around the world. And so it should be the same for Japan. But is that your view as well? And I am in a minority camp. And in that story, I think possibly a better option may be a nominal gross rate targeting instead of inflation targeting. One reason is that inflation comes from different sources, i.e. demand side contribution and suppressor contributions. But financial market doesn't care so much about the competition. And if supply-side factors push out prices, then financial

markets anticipate market position will respond by the interest rate. But this is actually harming the economy, especially for economy with such a low potential growth rate. So I think the nominal growth targeting can neutralize such an incentive for central bank to tighten monetary policy in response to a negative supply shock or either a positive or negative supply shock. And for example, if you see an oil shock, then your imports will rise, our nominal imports will rise, and that will actually dampen nominal growth so that if you target nominal GDP growth, you wouldn't.

raise interest rate in response to a rising oil prices. And that's probably more fair treatment or more balanced treatment between prices on the one hand and economic activity on the other hand. And globally, or at least among central banks in developed countries, 3% inflation targeting is forced upon to all countries, regardless of the potential growth. That's quite unfair to Japan and more harming for low potential growth country than high potential growth countries. So to avoid that shackle, I think for Japan it's probably better to focus on nominal GDP growth target by which you cannot compare Japan versus the rest of developed countries. And so that BOJ or Central Bank may be able to be a bit more independent in their system. Understood. And just out of interest, are there any BOJ members who are open to the idea of nominal GDP targeting? I am afraid not. Central Bank can determine which target they would get out. And about 10 or 15 years ago, they adopted 2% inflation target.

under Governor Clodagh. So that is kind of given to the BOJ. Although BOJ may be able to alter their target, but they have to consult with the government. And I'm not sure whether the government is either happy or not happy with the changing of the mandate. Understood. Now on the fiscal side, Japan is doing one of its... consumption tax changes. We've seen this many times before. Often it's a consumption tax hike. This time there's a proposal for a consumption tax cut on food. It's supposedly 8% to 1% by early next year. How are you thinking about this in terms of its impact on the distribution? Is it aggressive or not? And how will it impact the economy? When you lower the consumption tax, that we cannot raise it again, especially under a situation where we are on average family is still struggling to keep up with the rising crisis. And secondly, the consumption tax by nature is regressive. So that cutting tax on food would actually

benefit more the wealthy people than ordinary people. So that is not necessarily the objective of the government. But on that, is it not true that poorer people spend more on food as a share of income? So would it not be more helpful to them? However, in terms of the amount of spending in food as the total spending on food as percent of their income I think that the benefit was more on the high-income people. So it's probably not the ideal situation to give so much lidates to relatively wealthy or high-income people. And third is Japan-specific problem when you change the tax rate on consumption.

Then we've seen a huge dislocation of demand. In case of a consumption tax rate hike, we see a front-loading followed by a decline in demand. In the current case, we're going to see the reverse. And it is said to be lasting only for two years. So you would see two rounds of the opposite effects of consumption timing. the cut you see decline of spending and then after the cut you see a rush to spend. So and the reverse will happen when we raise tax rate in two years time. So it's a kind of more kind of more of a disruption to their connectivity. Although many people will be happier for those two years. And more broadly, another thing we've also been seeing, and you mentioned the poor and the rich, the distribution there. That said, I mean, we have seen an increase in wages occur. If you look at the numbers, you know, the spring updates as well. So at least from the outside, it does look like workers are earning more. And earlier you said, you know, there's a divergence between poor and rich people. But it is not the case that wages are going up. So the household should be feeling richer.

Yes, for this year, we are seeing per capita monthly wage growth on the order of 3% a year, I mean, scheduled wage growth. And this is the first time I think we saw such a wage growth of the level in probably 30 years. So it's the first time we see a slow but secular increase in wage growth. up to 3% a year. So that is finally giving some support for household income. Unfortunately, we've seen US war to push up the oil prices, which makes us a bit less happy. But aside from that, I think that household income conditions finally, but slowly getting better. In this particular year, government tries to lower prices of gas ring utilities and that will push down inflation by about half the percentage point. We'll give some real income growth for this year because of two things combined.

You mentioned the wage rises there. Some of the other work you've shown suggests that when it comes to the smaller companies and non-unionized workers, there's less scope to increase wages. Yes, when we look at the labor share or a cost of personnel as percentage of sales for companies by size of companies, we've seen those ratio coming down for larger companies, but not much mid-sized and smaller companies. In other words, the personal cost still is a relatively large proportion to sales, so that smaller companies can't afford to raise wages as much as larger companies. And when you look at the relative wages, say smaller company wages, divided by larger company wages. This ratio has not risen at all. At best, it's correct, which means the small companies try to raise wages relative to large peers, but they can't afford to do that at the moment.

And you know, another piece of work I've seen in some of your presentations is the hours worked of employees in Japan. You know, it's some of the charts you've shown, which we can also share later with listeners shows that hours worked has been declining quite sharply in Japan. And one would think in an aging population, you would actually increase hours of people, you know, because there's not as many people to work, but instead hours worked per person has been declining. Yes. That's a very worrying situation because the government intended to try to put a negative incentive against work in the form of work-style reforms since 2018 under the Abe Administration. What they have done is to encourage companies not to make people work so much over over time hours so that our labor input cannot go higher, regardless of the economic conditions. And there is one other constraint, especially for part-time workers, such that once the income, annual income goes up to some threshold, their disposal of income collapses because they start to pay social security tax. So those two are considered negative incentives for employees to work longer. And the similar pattern took place in 1990 at the height of the financial bubble. We lowered scheduled working hours a week from 48 hours to 40. And as a result, our hourly wages has shut up relative to labor productivity gains.

And that constraint acted as a negative supply shock and acted as an additional constraint for Japan's stagnation. I think similar situation replicated from 2018. We're going to change somehow, but it takes time. So despite the relatively benign economic condition, our working hours per

person still continue to fall. And you know, putting all of this together, I mean, what does this mean for the Bank of Japan? You know, of course, they started to raise interest rates. So, you know, do you think we could get to one and a half percent interest rates in the next 12 months? Or what's your sense of where the BOJ will go? The BOJ tries to be as flexible as possible so that when the situation for another round of interest rate hike seems they try to do it, but they try not to, you know, a certain interval for every interest rate rates. When they think it is a good timing, they may do it every three months. And as such, we feel they are flexible, but still quite aggressive. When they can, they will. That's my impression.

And there is a different aspect of the interest rate forecast. In my view, the short-term interest rate shouldn't be so high as the DOJ would anticipate. So there is a conflict between my should-be view and will-be view. But in terms of will-be view, I would imagine perhaps 125 basis point hike every six months. That's probably the best one. Understood. Now, you did earlier mention that the government has introduced various subsidies to reduce the cost of living issues for the Japanese. Presumably, that should have a fiscal impact. On top of that, the Takei-Aichi administration appears to be more stimulatory on the fiscal side. Do we need to worry about the fiscal picture in Japan? Of course, for many, many decades and how people have talked about the level of gross debt in Japan, even net debt. But how do you look at Japan's debt challenge? I think that the Takaiichi government is quite willing to deploy fiscal policy more often and more flexibly than previous administrations. In that sense, I am worried about sense of fiscal dominance over monetary policy. My fear is twofold, that on the Bank of Japan, they may be too late to raise rates so quickly. On the other hand, on the fiscal side, the government tries to dominate the Bank of Japan. So those two forces are getting stronger.

The second worry is that the government spending tends to be less efficient in relative to private sector decisions on spending. So the government can spend on items where the private sector demand doesn't exist. So it's an inefficiency as a long run. And regarding the fiscal condition, I think that the Japanese government conditions may not be as doomed as the financial market would anticipate. I have created my own fiscal vulnerability index. That's my own index, so you don't have to trust it. Consisting of fiscal side

variables like fiscal balance, government debt, net debt. and external side like current account balance and net external conditions, net external assets or abilities, interest payments as well. And combined those elements and evaluated that among 24 OECD-level countries, the balance vulnerability ranks in 16th, not the top. And the top was actually the US.

So if the vulnerability shows up, I think Japan will not be the first to blow up or be subject to the financial market pressure. Some other many other countries could go fast. That's my impression. Understood. And more broadly on the Takichi administration, how do you think the administration is doing so far? As you know, her approval rating has been quite high, although it is declining slowly. And her support base tends to be relatively younger generation, not older generation. So those younger generation may be a bit more fickle than others. mismanagement takes place then approval rating could start falling but so far it seems it has been still supported at elevated levels and perhaps a commitment to spend on many areas as a priority seems to be supported by the majority of Japanese people.

best on some of the ratings. Yeah, understood. And, you know, obviously one of the big themes in financial markets has been the weakness of the Japanese yen. It's been incredibly weak over the past year. Now, some of that has been attributed to the rise in all prices. So this is kind of a remnant of the Iran war. But even then, it appears to be even weaker than models would suggest. I mean, so do you have any explanations of why the yen has been so weak? I have to admit that. the asking question on currency on economists turn out to be very fruitless first and then I also look at several possible determinants which generate zero professional and some people talk about interest rate definitions that explain to some extent some other people look at the economic growth differential between say US and Japan And that explained to some extent as well. And some other people look at financial accounts of balance of payment like portfolio investment, direct investment, bank loans. And unfortunately, those portfolio investment doesn't explain so much. So I would imagine slight explanatory power in real interesting differential.

and economic growth differential, but not much on balance of payments or financial account side of the balance of payments. Yet we cannot explain

why yen is getting weaker and weaker despite reversal in real interest rate differential between us and Japan in the last 18 months. based on the person's probability, the end is perhaps four standard deviation weaker. So naturally, I think it's on the bottom right now. Yeah. In terms of interventions, the Japanese have intervened a number of times. Does that mean that from a policy perspective, the Japanese want a stronger currency? Or is it that they're worried that a weaker currency beyond current levels will somehow be the confidence issue for Japanese financial markets. I mean, what's their motivation for having intervention? The larger motive for intervention is probably to stop yen from depreciating quarter instead of making it stronger and stronger. Because beyond some certain levels of weaker yen, it could flip back to inflation in the sense that it caused a vicious cycle and negative shocks.

to any question. So I think that the Japanese government try not to reach that level again. But on the other hand, whether they are determined to appreciate the currency, I don't think so. Yeah, understood. Now before we go on to some of your more sort of global views, there is a piece of work I found interesting that you have done which is looking at Japanese birth rates and you know there's various incentive programs on childcare and so on which various governments and administrations in Japan have tried to use to boost birth rates but you've argued or your analysis seems to suggest work from home policies may actually be a better policy for boosting birth rates. Can you elaborate? When we look at very simply the correlation between total fertility rate and government spending on education and social security and childcare spending across developed countries. There has been virtually no correlation at all, which means no matter how much you spend to help people raise children and help one more children, the result seems to be very futile.

That's one conclusion. And secondly, there has been several recent articles on relationship between the work from home and for TTA. They suggest with one day per week from working from home, either male or female couple, then there are expected number of children, lifetime expected number of children. normalized by 0.3 to 0.5 kit. That quite substantially and statistically substantial increase. And that would mean if you allow some flexibility in the working arrangement for you or yourself or your

partner, then they may have a better chance to have the second or second kid. Government intervention tends to raise temporarily the birth rate, but this is only a transitory effect. Suppose you are 25 years old and you plan to have the first kid at 28 and second kid at 31, and you see the government policy support, so you started to think about the first kid at 26 and second at 29, but you don't plan the third one.

you just front loads the first kid that will push up the birth rate in second year or third year of the measure but it doesn't have a long run effect but if you have more certainty on work from home for longer say until the kid gets teenager then you can plan possibly the second or third child that maybe every year we haven't explored yet And in particular for Japan, where primary workers working from home share is very low, it's just one server, one quarter of some other developed countries here. So we still think, we think it may be, Japan may be able to get some next 40 to 80, thanks to more preparation from working from home.

Okay, great. On the global side, one of the big sort of themes over the last few years has been about the dollars reserve currency. You've done some work in this area as well. What are your findings? This is a very long-term story. And since the second Trump administration, I am worried about the US voluntarily gesturing their privileges of the reserve currencies by adopting unusual sets of trade policy and say defense policy and so on. And in my view, the reserve currency privilege is very important for non-resident investors, not U.S. domestic investors, but non-U.S. non-resident investors. So we look at their behavior on a casualty, look at their behavior and we find that for fixed income, their share of US asset class within a developed country fixed income asset. It is peaking in early part of this decade. For equities, it hasn't, because equity has a unique characteristic of platform industries totally concentrated in the US. They have to invest in the US because of that.

So equity share of non-resident investors pool of the US asset, that's not taking yet, but coming to a plateau at 50%. And lastly, when you look at the PE of equities between US and the rest of the Western developed countries, I found that The PE differential. USP is higher than other countries because of the US premium. But that differential started narrowing since the

beginning of 2015 when the second term administration started. That might suggest you cannot put so many larger premiums on PE on the US. The second Trump administration started very unusual and also docks set a background policy. That's my impression. And that if over the next two years and half, that now in trend of the P differential might continue. So far, I don't know. Yeah, understood. And you know, related to the tech story there, of course, there's been this big boom in AI and data center.

CAPEX, especially in the US, but also elsewhere as well. As an economist and looking at the business cycle, how do you interpret that in terms of its impact on the economy? It just does 0.3% increase. But now, in the last two years only, it has risen by 0.7 percentage point. It looks like a vertical jump in the last two years. So whether this exponential growth of spending on semiconductor shipments relative to the size of the economy can be sustained or not. I'm afraid to say no, because you have to cut back spending on the scale of 0.7% of GDP to allow for massive additional spending on same conductors. That is a great challenge for the global economy. And second, on the US, When we look at the CAPEX to GDP ratio, where you are denominated the CAPEX on machinery and equipment, including computers and pay-for-all equipment, and software and R&D investment, this share, as a reason, this share rose by about 3 percentage points from 1992 to 2000, was bursting with bubble, IT bubble.

by 3 percentage point and from 2009 after the bursting of the global finance crisis Towards first quarter this year. It has risen 2.9 percentage point from 8.4 percent to 11.6 percent but in terms of the expansion of the investment share as percent of GDP it has been a similar scale of increase to that of the IT boom period. So we are possibly approaching a relatively later stage of an expansion period, I would imagine. Although many people are still optimistic about AI and investment in data centers and so on. And the final question on the macro side is that the U.S. is also now facing a decline in the U.S. labor force. It's actually gone on for several decades, you know, for the last 40, 50 years. And Japan is more advanced in that in relation to that. I mean, so what's your view on that for the U.S.? Yes. When you look at the labor force participation rate across gender, across age, Japan's labor force participation rate has been sterilizing for all segments of the age and both genders. That's quite remarkable. And as a result, despite population aging,

where the share of elderly people in the population is getting bigger and bigger, our labor force participation rate is still rising. On the contrary, in the US, the labor force participation rate is collapsing, not because a particular age cohort participation rate is declining. It's more like a compositional change over the last 50 years. Every people share is getting larger. And because of that, US labor force participation rate is declining. I think that the US is suffering from the compositional change. Japan somehow managed to encourage people to remain in the workforce because we know we may live up to 100 years and we have to earn some income to some home so that we are forced to remain in the river market. But in the case of the US, some people may have become so financially affordable to exit the river market.

it's probably a very small fraction of those people. But generally, they are facing kind of secular drug, and they can't cope with in terms of the declining level of participants rate with the Asian. So it's quite poor, opposite outcome we had seen between those two countries. That's very interesting. So I mean, we've covered a lot of ground and that's been much appreciated from Japan to the US. Final question for you was if people wanted to connect with you, follow your work in some way, what's the best way for them to do that? I publish economic research at SVR securities. So one way to connect with me is to contact our salespeople or if you'd like to connect me directly.

you can have my email address in the document called the presentation document, and I probably can reply to several or fewer questions. Great. And people can use me as well as a link to you too. So with that, Matsuoka-san, it's been fantastic reconnecting with you. As usual, you have kind of big views, alternative takes as well, which is much appreciated. So thanks again. Thank you very much for your time. Bye-bye. MacroHive retains all ownership, title, rights and interests in this audio and video and all related content including audio transcripts, thumbnails and descriptions. You may share links to this publication and embed it using platform-native features. You must not, whether in whole or in part, download, copy, reproduce, redistribute, re-upload, edit, client or create derivative works from this content without the express written authority of MacroHive. This audio and video is made available to you for general information purposes only and

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